



Dhruv Milind Thakur

B.Tech Engineering Physics

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Degree	University/Institute	Year	CGPA/Marks(%)
B.Tech Engineering Physics	IIT Hyderabad	2028	7.6
XII (HSC Board Examinations in Maharashtra State)	Seth Hirachand Mutha college kalyan(west)	2024	76.00%
X (CBSE)	Seth Hirachand Mutha School kalyan(west)	2022	83.20%

POSITIONS OF RESPONSIBILITY

Head, Institute Technical Council - Operations Domain	April 2026
Assessment Lead, Office of Career Council , IIT Hyderabad	April 2026
Core, Management and Operation, Tinkerers' lab , IIT Hyderabad	June 2025 - May 2026
- Managed the Operational execution for procurement of Tinkerers' lab - Directed technical programs, conducted/executed events, and managed administration for the Thrust tech event.	

SKILLS

Programming: C/C++, JavaScript, Python

Frameworks & Libraries: Node.js, Fastify, Express.js, React.js, Pandas, NumPy, Scikit-learn, PyTorch

AI/LLM: LLMs, RAG, Embeddings, Qdrant, BM25, QLoRA, Ollama

Databases & Backend: PostgreSQL, MongoDB, Redis, REST APIs, WebSockets, JWT, Kafka

Tools & DevOps: Docker, Nginx, Git, GitHub Actions, AWS S3, Postman

RELEVANT COURSES

Computer Science & Mathematics: Introduction to Programming, Data Structures and Algorithms, Introduction to AI, Linear Algebra, Introduction to Probability, Introduction to Statistics, Basic Calculus

Physics & Engineering: Computational Physics, Quantum Mechanics, Analog Electronics, Digital Circuits.

Japanese Courses: Japanese Culture and Society

PROJECTS

High-Throughput Wallet & Ledger System Node.js (Fastify), PostgreSQL, Redis, Apache Kafka, Docker GitHub Reduced core transaction API latency by 98.10% (1526ms to 29ms), by offloading notification and logging tasks to asynchronous Apache Kafka consumer groups. Eliminated write deadlocks and balance drift under a load of 166+ TPS (throughput locally), by engineering a PostgreSQL double-entry ledger with alphabetical lexicographical row-locking (SELECT FOR UPDATE). Prevented double-spending and duplicate requests in under 1ms, by implementing a fast-path idempotency gate using Redis distributed locks (NX/PX keys) and write-through balance caches.
AI Financial Analyst Python (FastAPI), Qdrant, BM25, PyTorch, QLoRA, Ollama GitHub Evaluated an LLM financial analyst across 35 benchmark cases, achieving 100% execution success and tool-selection accuracy through schema-aware SQL generation, connection-level validation filters, and context controls. Designed a hybrid RAG pipeline (achieving 83.33% recall and 88.57% citations) and fine-tuned a local 1B model via QLoRA, boosting its task success rate from 40% to 91.43% (+51.43 pp) and reducing latency by 17%.
Financial Intelligence Platform Node.js, Python (Scikit-Learn, Pandas) GitHub Engineered a schema-compliant auditing and segmentation pipeline, verifying 1,800+ transactions for 100% balance correctness via sequential double-entry replays and using K-Means to cluster users into RFM cohorts. - Built a customizable ML engine and React dashboard using scikit-learn Isolation Forest (isolating 3% outliers) and Recursive Random Forest (94% R² accuracy) to project 30-day platform cashflows.
MapLink React.js, Node.js (Express), Socket.io, PostgreSQL, PostGIS, Redis, Nginx, Docker, GitHub Actions GitHub - Reduced database write transactions by 98% and query latency by 97% for historical routes by implementing a Redis Write-Behind Cache (15s bulk flusher) combined with PostgreSQL composite and GiST spatial indexes - Reduced client location requests by 75% using a Haversine filter with a 3s update interval and 5m threshold - Designed a horizontally scaled backend cluster with 3 Node.js instances load-balanced by Edge Nginx (sticky ip_hash), integrating a Redis Pub/Sub Adapter to sync WebSocket broadcasts with an 8ms cross-server latency.

EXTRACURRICULAR

Sports - Playing football under NSO IIT Hyderabad

[Leetcode](#) - Solved **500+** **Data Structures and Algorithms** problems on LeetCode

EXTRA COURSES

Introduction to Generative AI | Google Cloud | [Certificate](#)

Introduction to Prompt Engineering with GitHub Copilot | Microsoft | [Certificate](#)

Get Started with Databricks for Generative AI | Databricks | [Certificate](#)

OPEN SOURCE CONTRIBUTION

Contributed to the Supabase Auth Go backend by implementing exact, case-insensitive user lookup by email on the admin API; bypassed slow fuzzy searches with index-supported PostgreSQL queries and verified functionality using Go unit testing and Docker environments to support client SDKs. | [PR #2668](#)

Contributed to Rakuten Tech's QueryCraft AI-SQL assistant by implementing offline Prisma ORM schema parsing; designed a TypeScript transpiler to convert Prisma models into SQL DDL for the LLM context, bypassed Next.js API credential validations, and verified compilation using TypeScript. | [PR #109](#)

TECHNICAL & CULTURAL INTERESTS

System Harmony (*Wa*): Analyzing the algorithmic efficiency and high-throughput routing of Japanese transit logistics to inspire deadlock-free, distributed backend architectures.

Proactive Architecture (*Omotenashi*): Exploring how the Japanese spirit of hospitality translates into frictionless UX, driving the design of zero-latency caches and user-centric software.